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Open frontier foundation models: open pipelines for sovereign frontier ML & AI

Jülich Supercomputing Center (JSC); FZJ

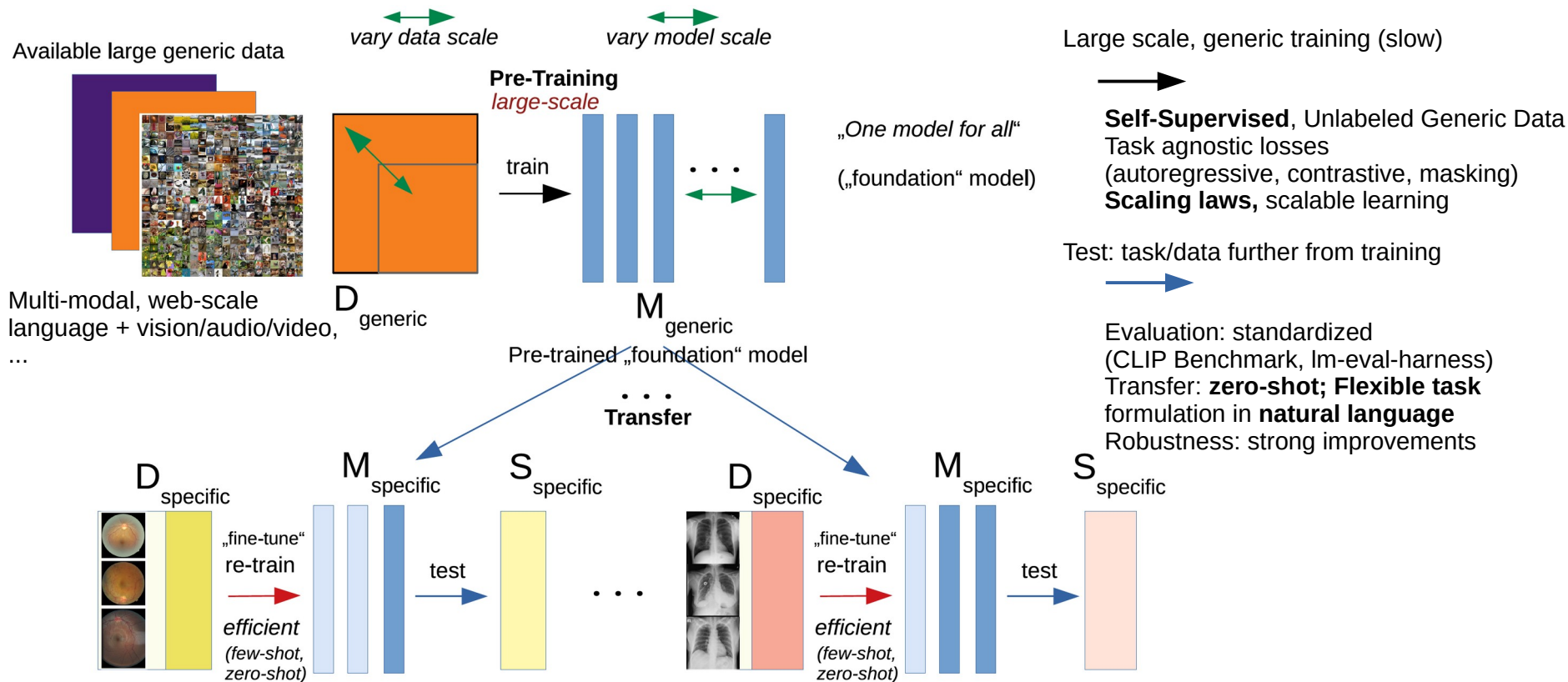
Scalable Learning & Multi-Purpose AI Lab (SLAMPAI)

Large-scale Artificial Intelligence Open Network (LAION)

European Laboratory for Learning and Intelligent Systems (ELLIS)

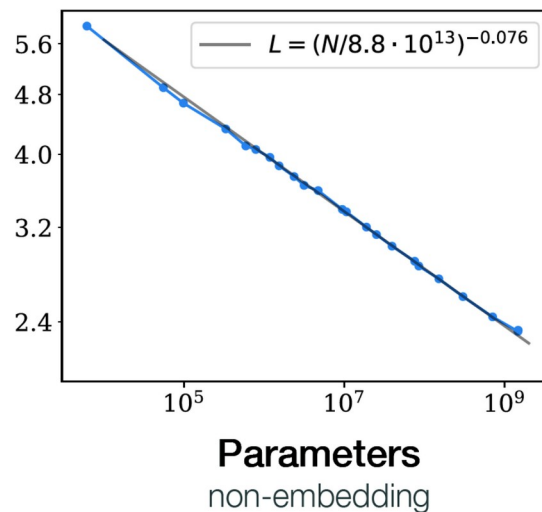
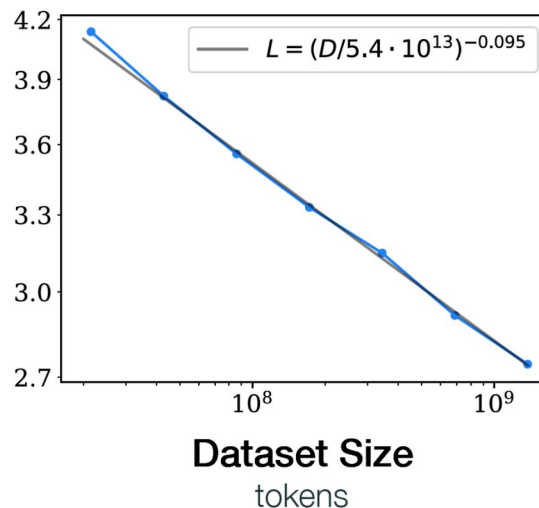
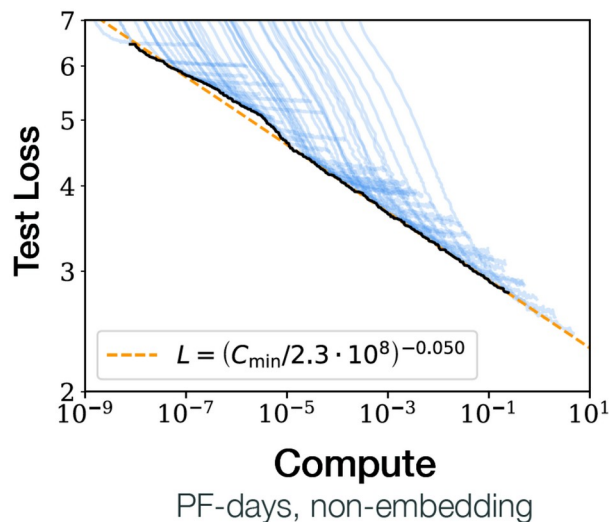
Foundation models: generic transferable learning

- Core breakthrough: **generalist learning that transfers across conditions/tasks**



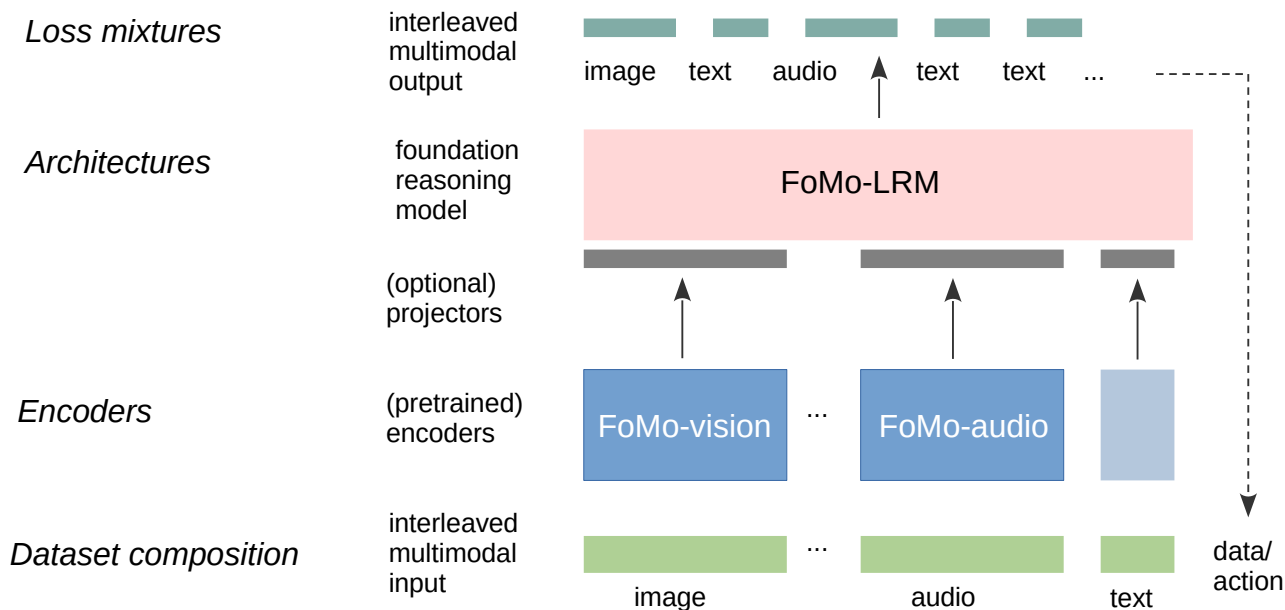
Foundation models: generic transferable learning

- **Strong transfer:** zero-shot, few-shot on variety of tasks in new conditions
 - Novel tasks can be defined in-context using natural language
- **Scaling laws:** predictable function and transfer improvement with scale (compute, model, data)



Foundation models: generic transferable learning

- **Downstream re-usage:** pre-trained components re-used in further downstream systems, eg multi-modal generalist foundation models
 - no or only little adaptation required
 - Qwen 3.5 VL/Audio/Omni, Emu 3.5, pi-0 : agentic and world models



Research communities for open foundation models

- Problem – building, studying & **improving** foundation models is challenging. Requires:
 - **large-scale data** (at least 100M of samples; > 100B of tokens)
 - **large-scale compute** (GPU years per single experiment)
 - **expertise** in large-scale machine learning (eg., distributed training) and scaling law measurement from hundreds experiments at smaller scales
 - Broad research community cut off from training & studying strong transferable models at various (especially larger) scales

→ Solution



Registered as **non-profit research LAION e.V.** since 2021 in Hamburg

Open foundation models & datasets

- Making **entire pipeline** – dataset composition, model training, benchmarks & evaluation – **fully reproducible & modifiable**

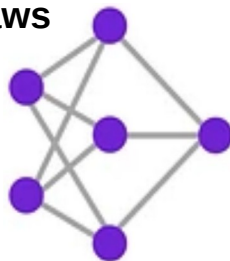
OPEN-SOURCE

Dataset &
Dataset composition



OPEN-SOURCE

Training procedure,
model weights,
checkpoints, **scaling
laws**



OPEN-SOURCE

Evaluation benchmarks,
downstream transfer procedures



Pre-training. Supercomputers for experiment execution required!

Re-LAION-5B/DataComp-1B
LAION-audio-630k
DCLM-baselines
OpenThoughts

openCLIP/openMammut
LAION-CLAP
DCLM
OpenThinker

openCLIP Benchmark,
CLAP benchmark,
Lm-eval-harness, Evalchemy,
Terminal-Bench, Harbor

<https://github.com/mlfoundations/datacomp/>

https://github.com/mlfoundations/open_clip

https://github.com/LAION-AI/CLIP_benchmark/



Open foundation models & datasets

- Making **entire pipeline fully open**: crucial for collaborative research

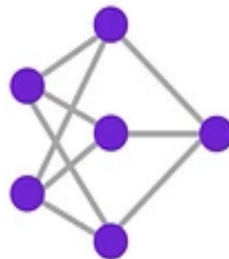
OPEN-SOURCE

Dataset interventions



OPEN-SOURCE

Training procedure interventions



OPEN-SOURCE

Standardized evaluation



- Standardizing pipeline: improving parts and whole procedure in common action, comparison via scaling laws derived in common controlled frame
- Collaborative search for interventions that lead to stronger scalable learning procedures



Open foundation models: improving scaling

- Interventions along the **whole reproducible pipeline** – dataset composition, model training, benchmarks & evaluation – to boost scaling

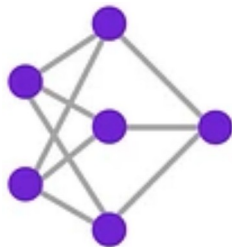
OPEN-SOURCE

Dataset &
Dataset composition



OPEN-SOURCE

Training procedure,
model weights,
checkpoints



OPEN-SOURCE

Evaluation benchmarks,
downstream transfer procedures



Supercomputers required!

Dataset
composition
studies, scaling
laws

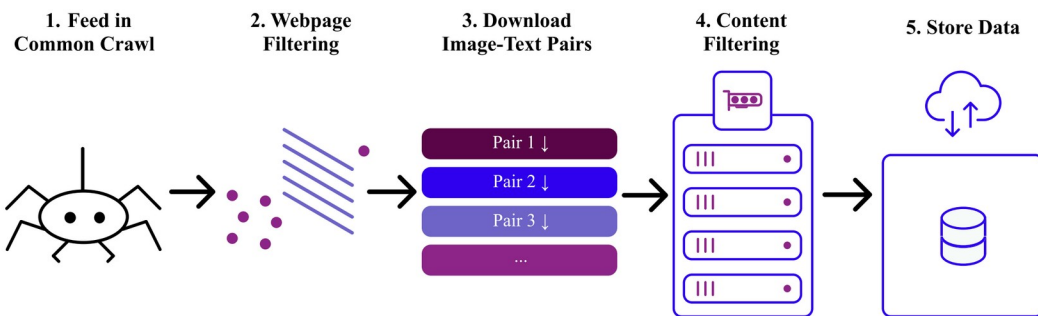
Learning
procedure
studies, scaling
laws

Benchmarks to
measure
generalization,
transfer capability



Open large-scale reference/foundation data

- LAION-400M/5B: Open sourcing data collection procedures - transparent dataset, open source tooling for reproducible dataset composition (**NeurIPS Outstanding Paper Award 2022**)
- Open dataset: collection of text and links to images on public Internet

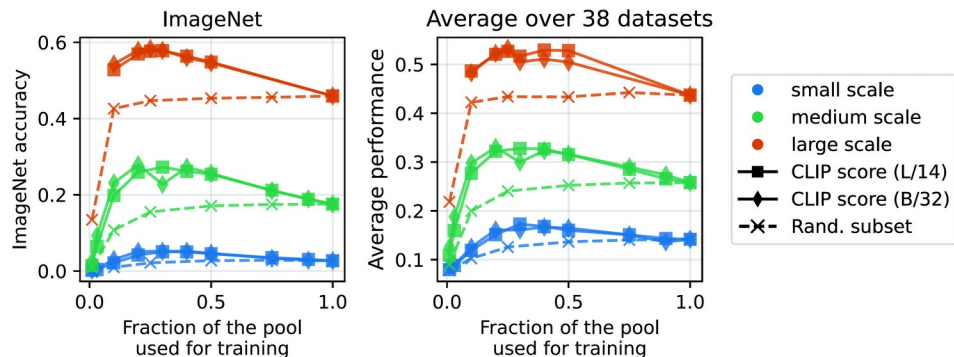
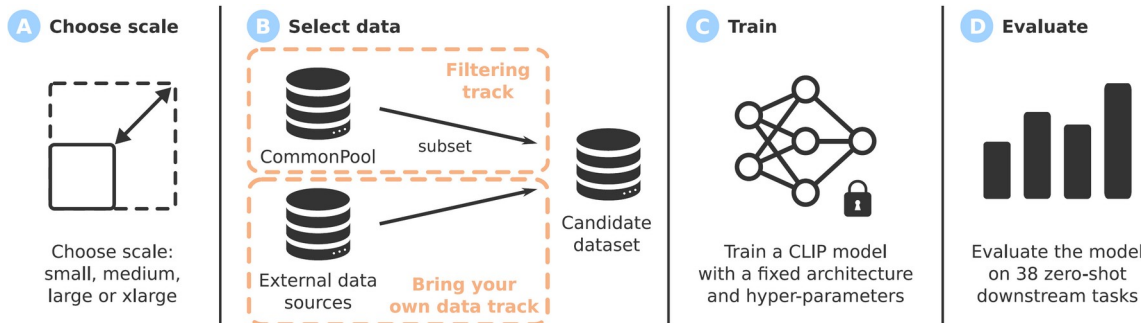


Dataset	# English Img-Text Pairs
Public Datasets	
MS-COCO	330K
CC3M	3M
Visual Genome	5.4M
WIT	5.5M
CC12M	12M
RedCaps	12M
YFCC100M	100M ²
LAION-5B (Ours)	2.3B
Private Datasets	
CLIP WIT (OpenAI)	400M
ALIGN	1.8B
BASIC	6.6B



Data-centric scaling law interventions

- DataComp, DataComp-LM: what constitutes good data for FoMo training?

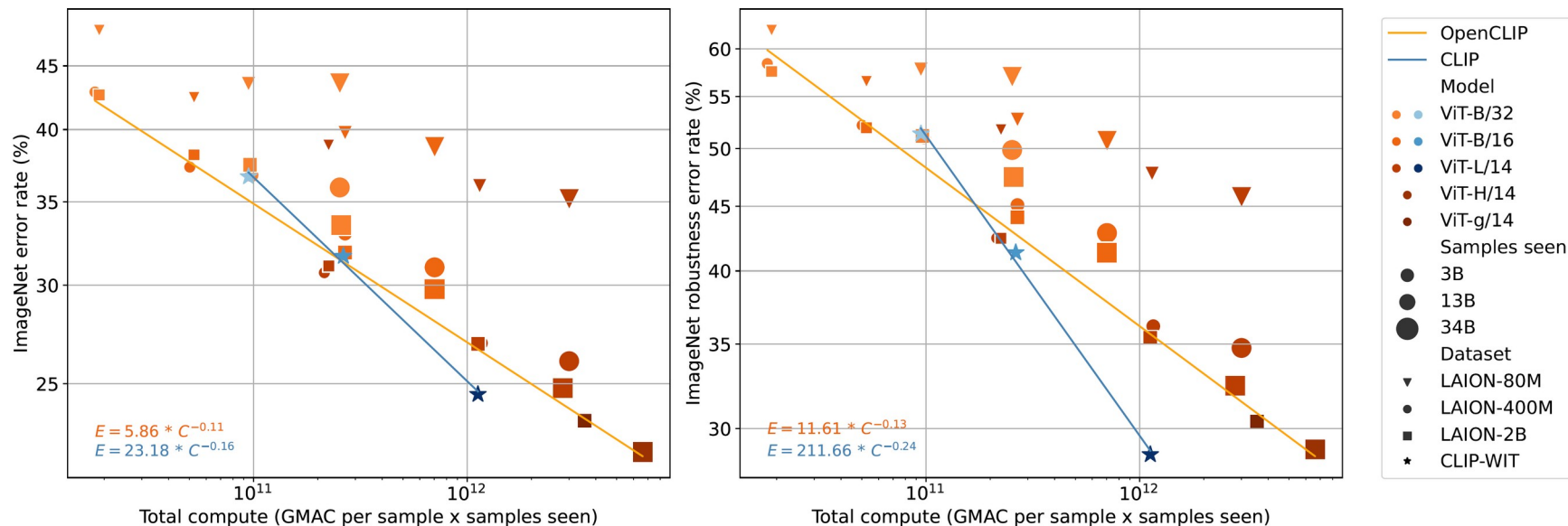


Dataset	Dataset size	# samples seen	Architecture	Train compute (MACs)	ImageNet accuracy
OpenAI's WIT [111]	0.4B	13B	ViT-L/14	1.1×10^{21}	75.5
LAION-400M [128, 28]	0.4B	13B	ViT-L/14	1.1×10^{21}	72.8
LAION-2B [129, 28]	2.3B	13B	ViT-L/14	1.1×10^{21}	73.1
LAION-2B [129, 28]	2.3B	34B	ViT-H/14	6.5×10^{21}	78.0
LAION-2B [129, 28]	2.3B	34B	ViT-g/14	9.9×10^{21}	78.5
DATAComp-1B (ours)	1.4B	13B	ViT-L/14	1.1×10^{21}	79.2



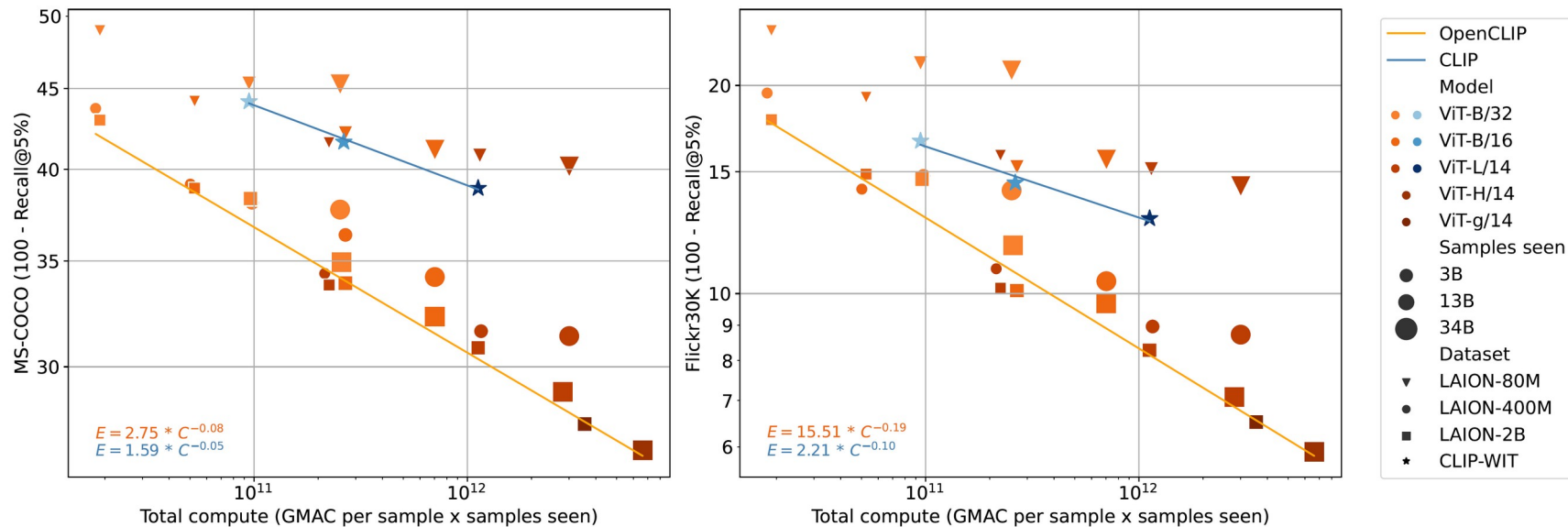
Scaling laws for open foundation models

- Scaling laws with LAION-400M/2B and openCLIP: open-source data, models and code - **reproducible** science of foundation models
- Below: zero-shot image classification, ImageNet-1k & robustness sets



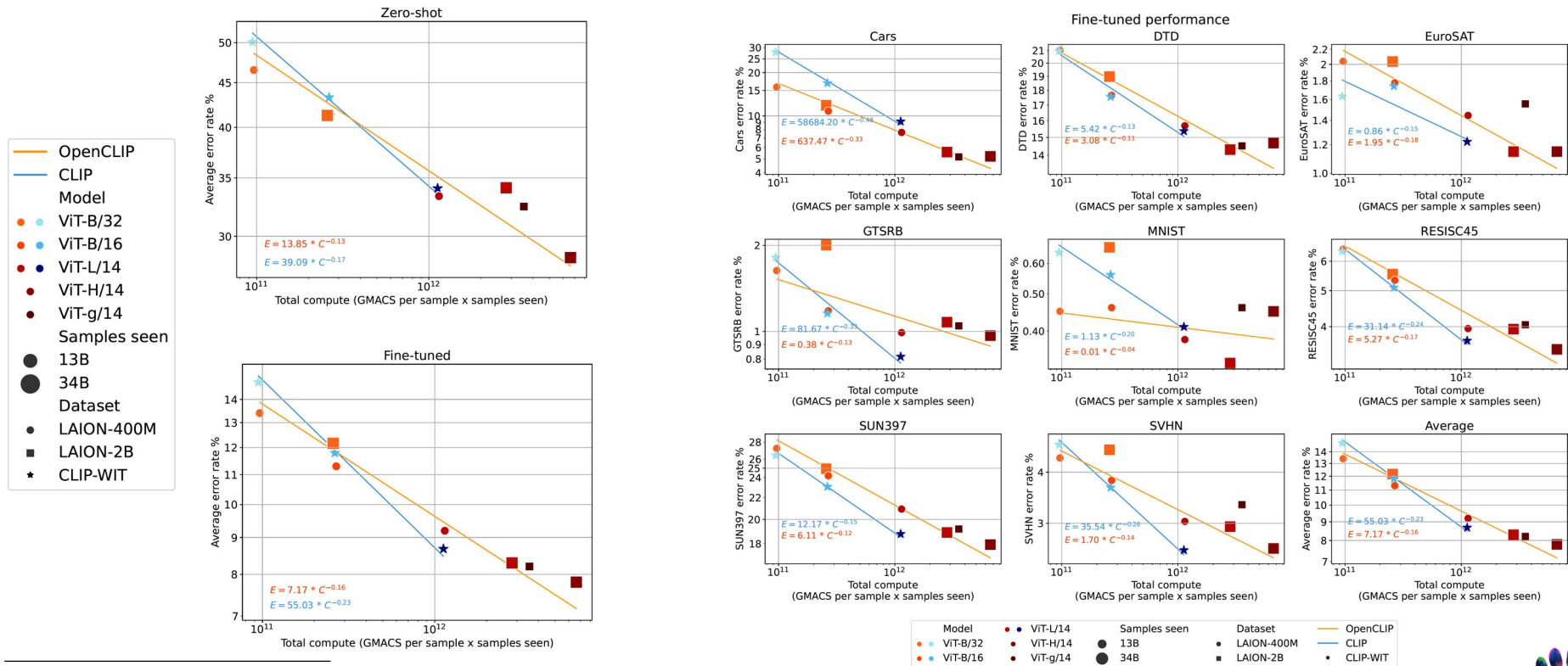
Scaling laws for open foundation models

- Comparing LAION-400M/2B (LAION) and WIT (openAI)
- **Matching or outperforming strong closed models** by using open data
 - LAION as a **open frontier lab**: building **open** foundation models that match **strongest** state-of-the-art from closed industry labs



Scaling laws for open foundation models

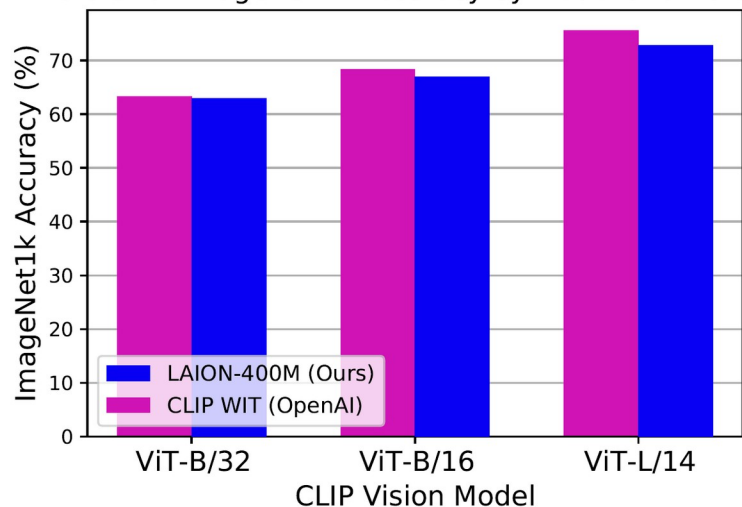
- Scaling laws for various transfer procedures and downstream datasets



Open frontier foundation models & datasets

- **Predictably outperforming strong closed models** by using open data
- LAION as an **open frontier lab**: building **open** foundation models that match **strongest** state-of-the-art from closed industry labs
- Grounds for further strong open-source research – eg Stable Diffusion, DataComp

Zero-Shot ImageNet1k Accuracy by Model and Dataset



Dataset	# English Img-Txt Pairs
Public Datasets	
MS-COCO	330K
CC3M	3M
Visual Genome	5.4M
WIT	5.5M
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Private Datasets	
CLIP WIT (OpenAI)	400M
ALIGN	1.8B
BASIC	6.6B



Outstanding Paper



Open frontier foundation models & datasets

- Open-source releases: > 100M of downloads for pre-trained openCLIP models; >10k stars for code repository; same success for CLAP (audio)

OpenCLIP DataComp

OpenCLIP LAION-2B

CLAP: Contrastive Language-Audio
Pretraining



OpenCLIP LAION-2B

OpenCLIP models trained on LAION-2B

🐾 laion/CLIP-ViT-bigG-14-laion2B-39B-b160k
Zero-Shot Image Classification • Updated Jan 16 • 📄 415k • ❤️ 226

🐾 laion/CLIP-ViT-g-14-laion2B-s34B-b88K
Zero-Shot Image Classification • Updated Mar 22 • 📄 13.7k • ❤️ 18

🐾 laion/CLIP-ViT-g-14-laion2B-s12B-b42K
Updated Feb 23 • 📄 38.2k • ❤️ 39

🐾 laion/CLIP-ViT-H-14-laion2B-s32B-b79K
Zero-Shot Image Classification • Updated Jan 16 • 📄 973k • ❤️ 305

🐾 laion/CLIP-ViT-L-14-laion2B-s32B-b82K
Zero-Shot Image Classification • Updated Jan 16 • 📄 80k • ❤️ 43

🐾 laion/CLIP-ViT-B-16-laion2B-s34B-b88K
Zero-Shot Image Classification • Updated Apr 19, 2023 • 📄 5.81M • ❤️ 27

🐾 laion/CLIP-ViT-B-32-laion2B-s34B-b79K
Zero-Shot Image Classification • Updated Jan 15 • 📄 1.58M • ❤️ 89

GitHub repository view for `mlfoundations / open_clip`. The repository is public and has 76 issues, 35 pull requests, and 47 tags. It is a Python project for an open source implementation of CLIP.

Repository structure:

File	Description	Last commit
<code>.github/workflows</code>	Add build to deploy pip installs	last month
<code>docs</code>	Refactor build / dist to use pyproject.toml (#909)	last month
<code>scripts</code>	Refactor build / dist to use pyproject.toml (#909)	last month
<code>src</code>	Release 2.26.1	last month
<code>tests</code>	Refactor build / dist to use pyproject.toml (#909)	last month
<code>tutorials</code>	Quick fixes for int8 inference, as well as tutorial (#508)	last year

Tags: `computer-vision`, `deep-learning`, `pytorch`, `pretrained-models`, `language-model`, `contrastive-loss`, `multi-modal-learning`, `zero-shot-classification`.



Open foundation models: improving scaling

- Interventions along the **whole reproducible pipeline** – dataset composition, model training, benchmarks & evaluation – to boost scaling

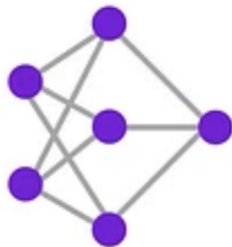
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Supercomputers required!

Dataset
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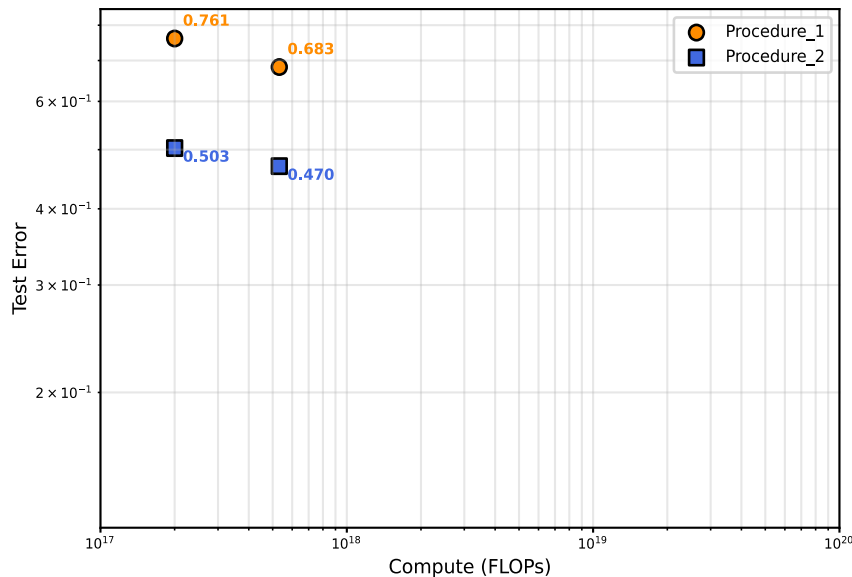
Benchmarks to
measure
generalization,
transfer capability



Improving foundation models: comparison

- How to determine which learning procedure leads to better foundation models: what interventions matter, which procedure is worth scaling up?

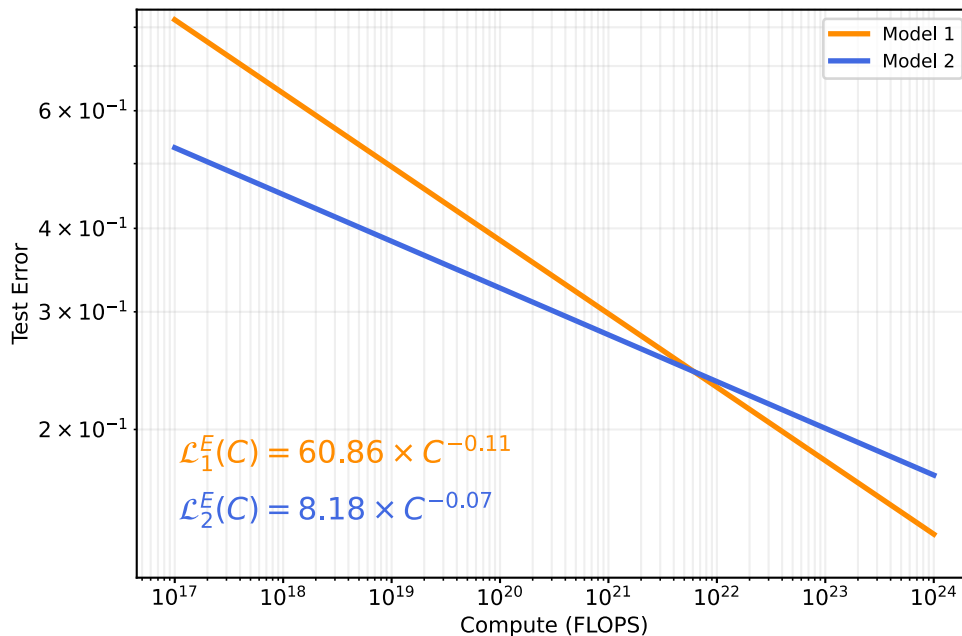
Procedure	Arch	Params	Dataset	Samples Seen	Compute (FLOPs)	Test Error
Procedure 1	Arch 1	0.5B	Dataset X	30M	2.00e+17	0.761
	Arch 1	0.5B	Dataset X	80M	5.32e+17	0.683
Procedure 2	Arch 2	0.5B	Dataset X	30M	2.00e+17	0.503
	Arch 2	0.5B	Dataset X	80M	5.32e+17	0.470



Scaling laws: learning procedure comparison

- Comparison using single isolated points can be highly misleading

Procedure	Arch	Params	Dataset	Samples Seen	Compute (FLOPs)	Test Error
Procedure 1	Arch 1	0.5B	Dataset X	30M	2.00e+17	0.761
	Arch 1	0.5B	Dataset X	80M	5.32e+17	0.683
Procedure 2	Arch 2	0.5B	Dataset X	30M	2.00e+17	0.503
	Arch 2	0.5B	Dataset X	80M	5.32e+17	0.470



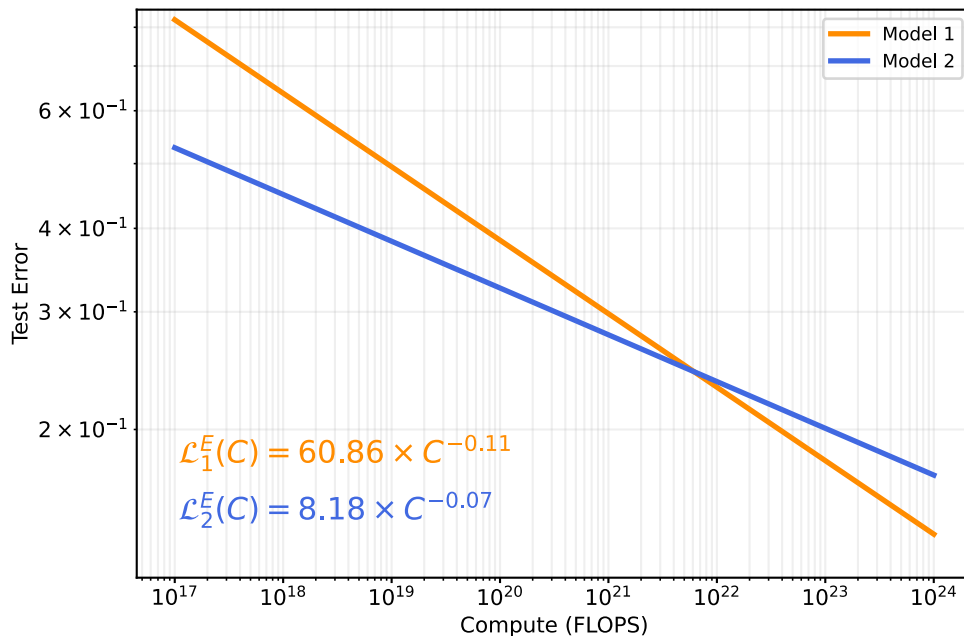
$$\mathcal{L}(C) = C_c \cdot C^{-\alpha_C} + L_\epsilon$$



Scaling laws: learning procedure comparison

- Comparison requires scaling law derivation using standardized open procedures
 - measuring scaling span instead a single reference point, predicting scaling up
 - conducting by fully controlling dataset composition, training, transfer/evals

$$\mathcal{L}(C) = C_c \cdot C^{-\alpha C} + L_\epsilon$$



- Learning procedure 1 vs Learning procedure 2
- Scenarios:
 - Comparing Model 1 vs Model 1 while fixing same open data
 - Comparing open Dataset 1 vs Dataset 2 while fixing same open training procedure / model arch
 - ...



Search for stronger scalable foundation models

- Comparing foundation models/datasets via scaling law derivation using open pipelines (CLIP vs. MaMMUT; open datasets - DataComp, Re-LAION, DFN)

Scaling Laws for Robust Comparison of Open Foundation Language-Vision Models and Datasets

Marianna Nezhurina^{1,2,5§*} Tomer Porian^{1,2,5*} Giovanni Pucceti³ Tommie Keressies^{1,4}
Romain Beaumont¹ Mehdi Cherti^{1,2,5§*} Jenia Jitsev^{1,2,5*}

¹LAION ²Juelich Supercomputing Center (JSC), Research Center Juelich (FZJ)

³Institute of Information Science and Technologies “A. Faedo” - CNR Pisa

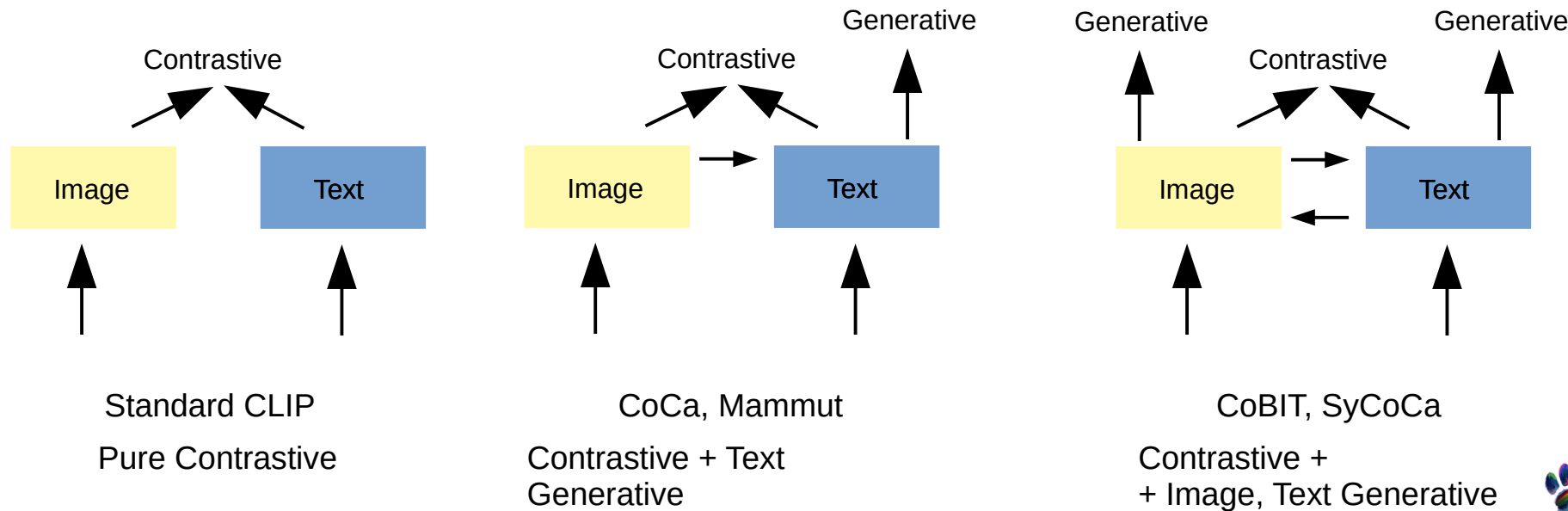
⁴Eindhoven University of Technology

⁵Open-Ψ (Open-Sci) Collective



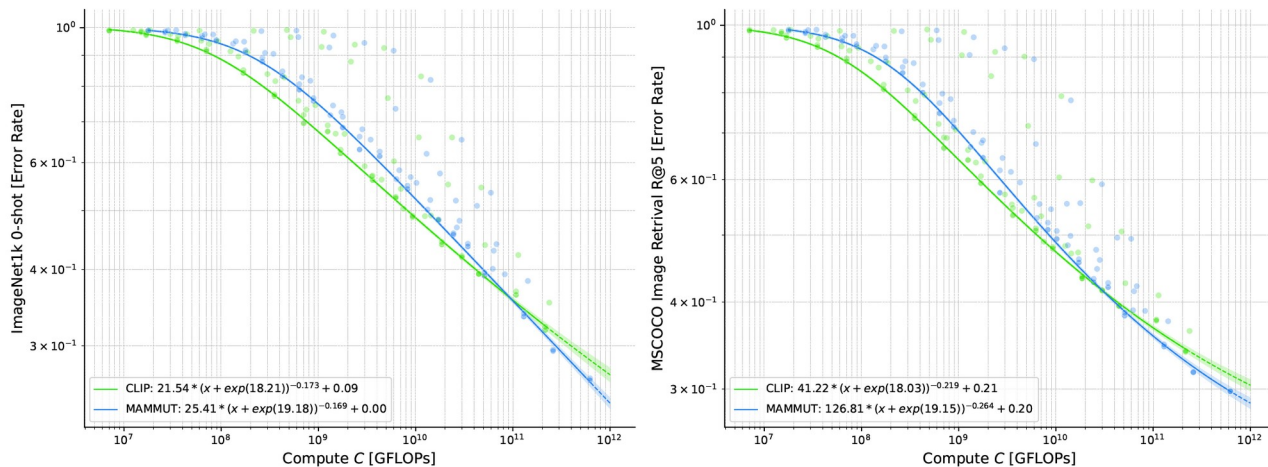
Search for stronger scalable foundation models

- Re-LAION, DataComp & DFN: improving datasets for pre-training
- OpenCLIP extensions: improving learning procedure
 - extend for text & image generative losses (CoCa, Mammut)
 - what loss mix might have stronger scaling?



Scaling laws: learning procedure comparison

- Comparison via scaling law derivation: predicting which procedure to prefer



(a) ImageNet-1k 0-shot classification

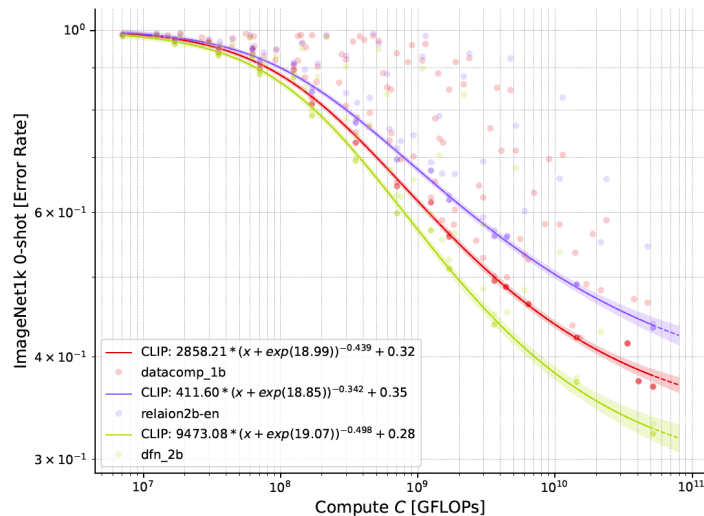
(b) MS-COCO image R@5

Figure 2: **Scaling on Re-LAION-1.4B.** Comparison of CLIP and MaMMUT via scaling laws on Re-LAION-1.4B. Error rate on downstream tasks is plotted against compute. MaMMUT outperforms CLIP in terms of scalability, indicated by crossing scaling law fit lines, where MaMMUT takes over CLIP in performance from larger compute scale $> 10^{11}$ GFLOPs on, showing similar trends as on DataComp-1.4B.

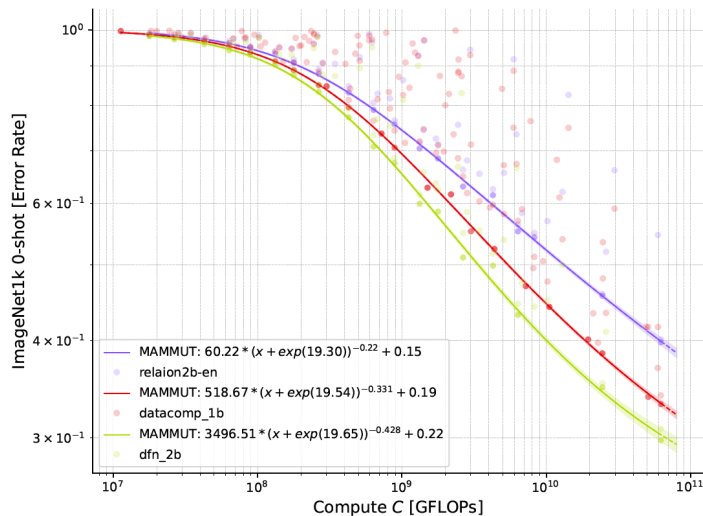


Scaling laws: learning procedure comparison

- Comparison via scaling law derivation: open dataset comparison



(a) IN-1k 0-shot error rate for openCLIP



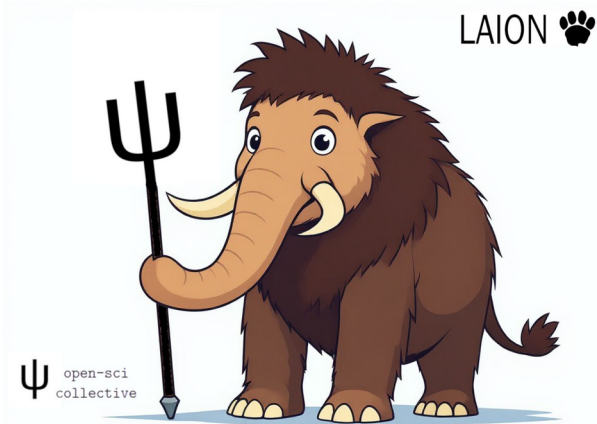
(b) IN-1k 0-shot error rate for openMaMMUT

Figure 6: Scaling laws for IN1k 0-shot performance of openCLIP (left) and openMaMMUT (right), comparing training on Re-LAION-1.4B, DataComp-1.4B and DFN-1.4B. Training on DFN-1.4B results in superior performance across scales consistently for both architectures.



Open foundation models with stronger scalability

- LAION as open frontiers lab: openMaMMUT predictably matching or outperforming SOTA of closed labs
- Scaling law based comparison: predicting whether an experimental procedure is worth scaling up, leading to stronger models than an already existing reference



ViT	Res.	Seq.	Model	Dataset	#Samples	ImageNet-1k		COCO	
						val	v2	T→I	I→T
L/16	256	256	SigLIP [18]	WebLI-10B	40B	80.44	73.76	75.26	88.40
			SigLIP 2 [14]	WebLI-10B	40B	<u>82.35</u>	<u>76.66</u>	<u>76.84</u>	<u>90.44</u>
L/14	224	256	OpenCLIP [10]	LAION-2B	34B	75.24	67.73	70.46	84.30
			CLIP [7]	WIT-400M	12.8B	75.54	69.84	59.95	79.56
			MetaCLIP [45]	MetaCLIP-2.5B	12.8B	79.19	72.64	<u>71.36</u>	84.94
			EVA-CLIP [46]	Merged-2B	4B*	79.75*	72.92*	70.68	85.26
			DFN [20]	DFN-2B	13B	81.41*	74.58*	73.19*	86.20*
			DataComp [19]	DataComp-1.4B	12.8B	79.19	72.06	69.86	84.64
			OpenMaMMUT (Ours)	DataComp-1.4B	12.8B	<u>80.34</u>	<u>73.78</u>	71.19	<u>85.88</u>

Table 3: Zero-shot classification (accuracy) and retrieval (R@5) results. DFN used ImageNet/MS-COCO-finetuned model for data filtering; EVA-CLIP was initialized from models pre-trained on ImageNet. We use **bold** for best overall results, gray for models involving ImageNet/MS-COCO data as training data in pipeline, and underlined for best results without ImageNet/MS-COCO involvement.



Open foundation models & datasets

- DataComp-LM: fully open, reproducible pipeline for language modelling; fully open data (DCLM-Baseline, 4.4T tokens in total) & models (DCLM-1B/7B); predictably match/outperform SOTA models (eg Llama-3-8B)

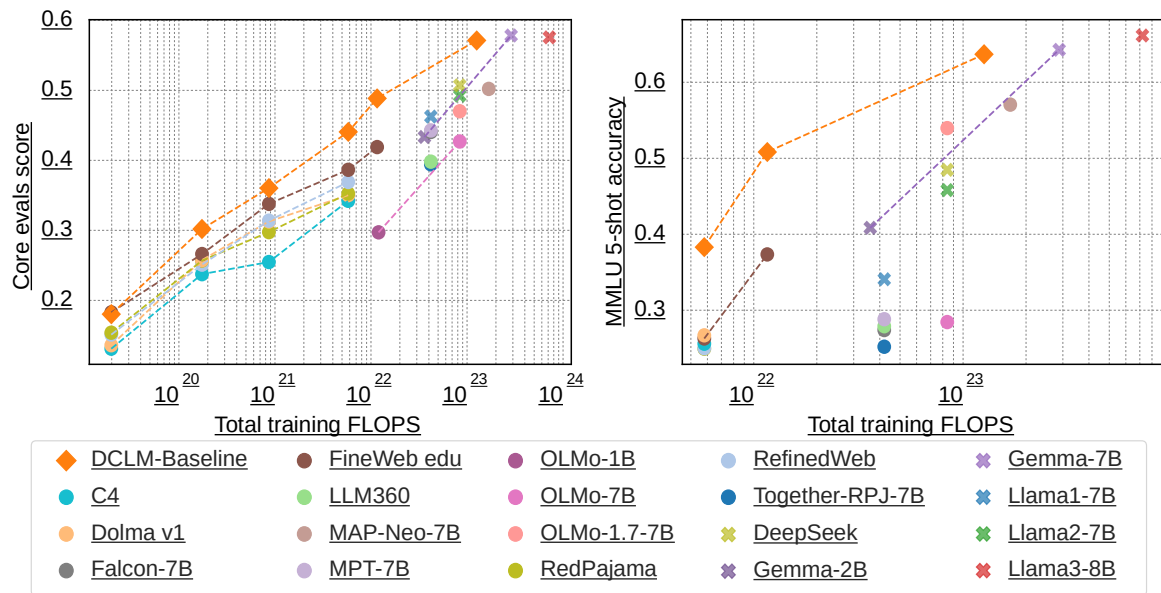
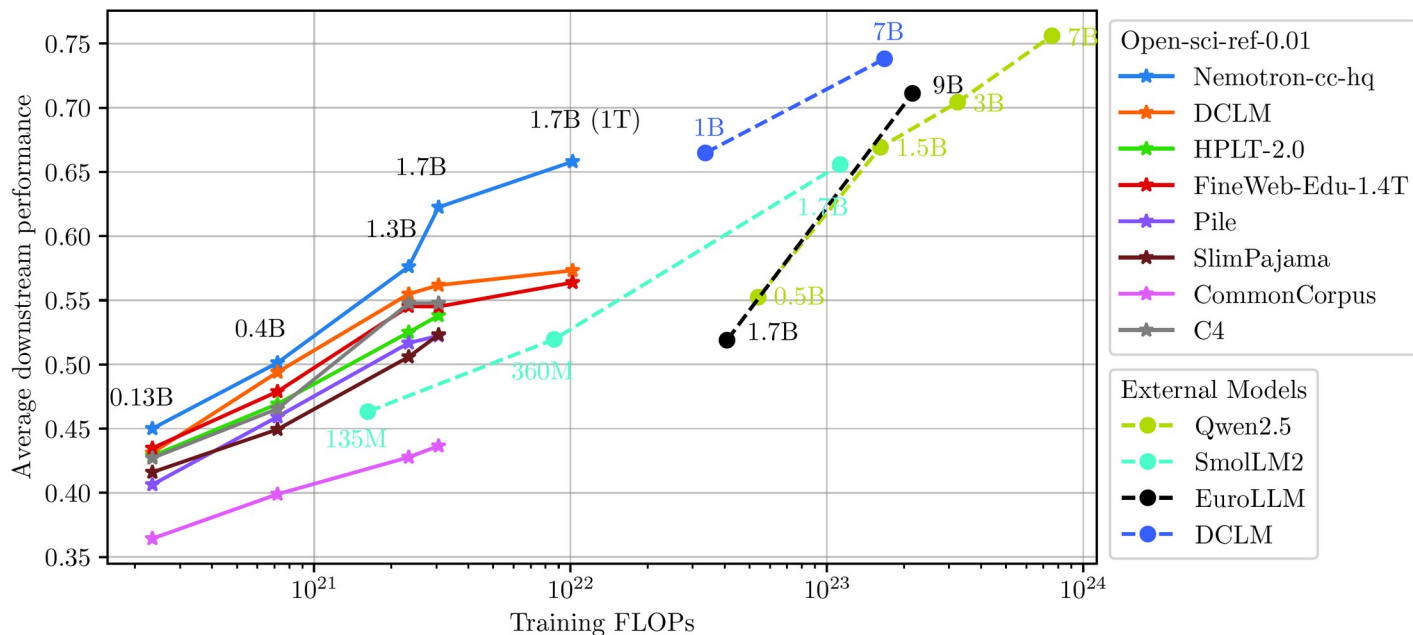


Figure 1: **Improving training sets leads to better models that are cheaper to train.**



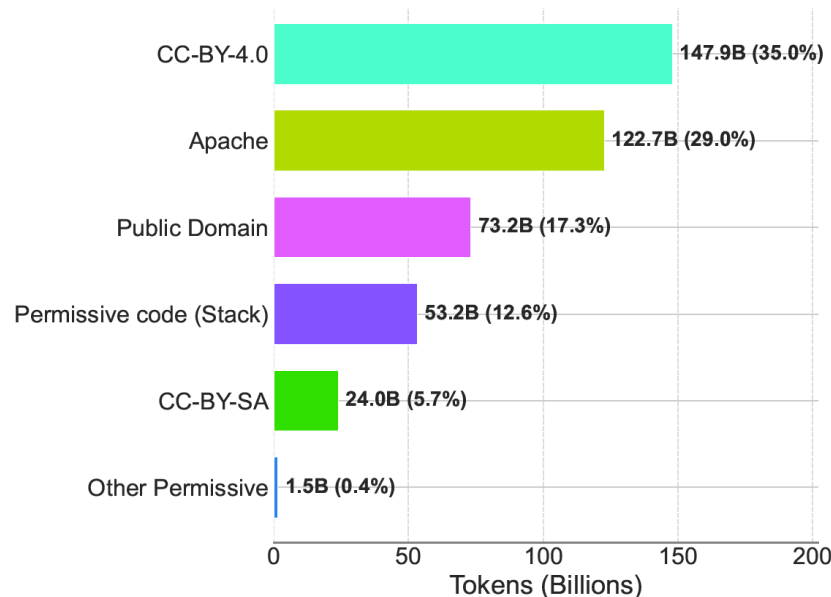
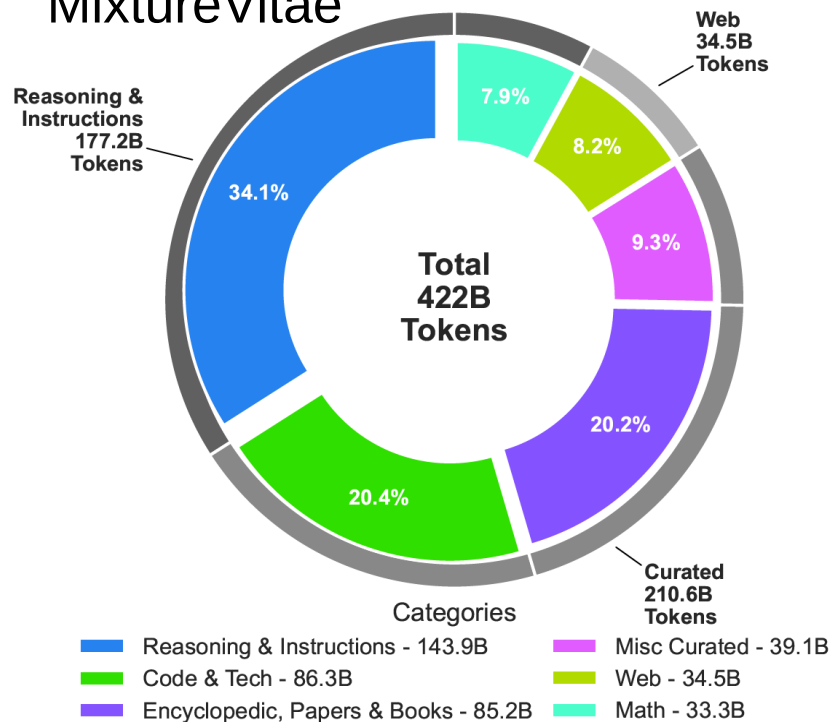
Open foundation models & datasets

- Open-sci-ref-0.01 : comparison in common frame on aligned compute



Open pipeline: permissive data & training

- Open permissive dataset for incepting strong reasoning in pre-training: MixtureVITae



open-sci
collective



Open pipeline: reasoning post-training



Open Thoughts

DATA RECIPES FOR REASONING MODELS

**Etash Guha^{*1,2}, Ryan Marten^{*3}, Sedrick Keh^{*4}, Negin Raoof^{*5}, Georgios Smyrnis^{*6},
Hritik Bansal^{ζ7}, Marianna Nezhurina^{ζ8,9,16}, Jean Mercat^{ζ4}, Trung Vu^{ζ3}, Zayne Sprague^{ζ6},
Ashima Suvarna⁷, Benjamin Feuer¹⁰, Liangyu Chen¹, Zaid Khan¹¹, Eric Frankel²,
Sachin Grover¹², Caroline Choi¹, Niklas Muennighoff¹, Shiye Su¹, Wanjia Zhao¹, John Yang¹,
Shreyas Pimpalgaonkar³, Kartik Sharma³, Charlie Cheng-Jie Ji³, Yichuan Deng²,
Sarah Pratt², Vivek Ramanujan², Jon Saad-Falcon¹, Jeffrey Li², Achal Dave, Alon Albalak¹³,
Kushal Arora⁴, Blake Wulfe⁴, Chinmay Hegde¹⁰, Greg Durrett⁶, Sewoong Oh²,
Mohit Bansal¹¹, Saadia Gabriel⁷, Aditya Grover⁷, Kai-Wei Chang⁷, Vaishaal Shankar,
Aaron Gokaslan¹⁴, Mike A. Merrill¹, Tatsunori Hashimoto¹, Yejin Choi¹,
Jenia Jitsev^{8,9,16}, Reinhard Heckel¹⁵, Maheswaran Sathiamoorthy³,
Alexandros G. Dimakis^{†3,5}, Ludwig Schmidt^{†1}**

¹Stanford University, ²University of Washington, ³BespokeLabs.ai, ⁴Toyota Research Institute,

⁵UC Berkeley, ⁶UT Austin, ⁷UCLA, ⁸JSC, ⁹LAION, ¹⁰NYU, ¹¹UNC Chapel Hill,

¹²ASU, ¹³Lila Sciences, ¹⁴Cornell Tech ¹⁵TUM ¹⁶Open-Ψ (Open-Sci) Collective



Open pipeline: reasoning post-training

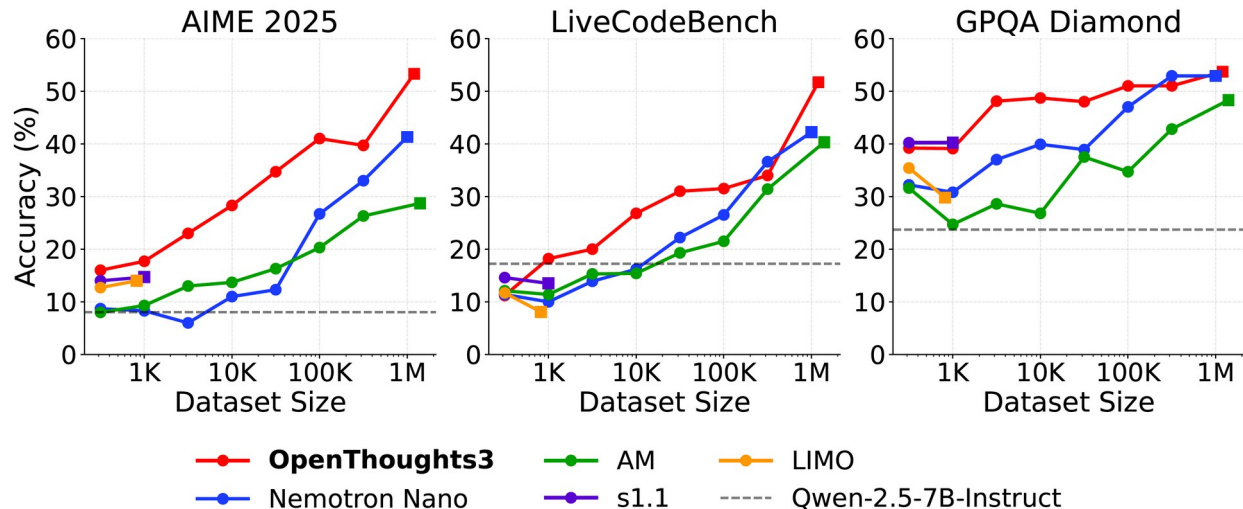
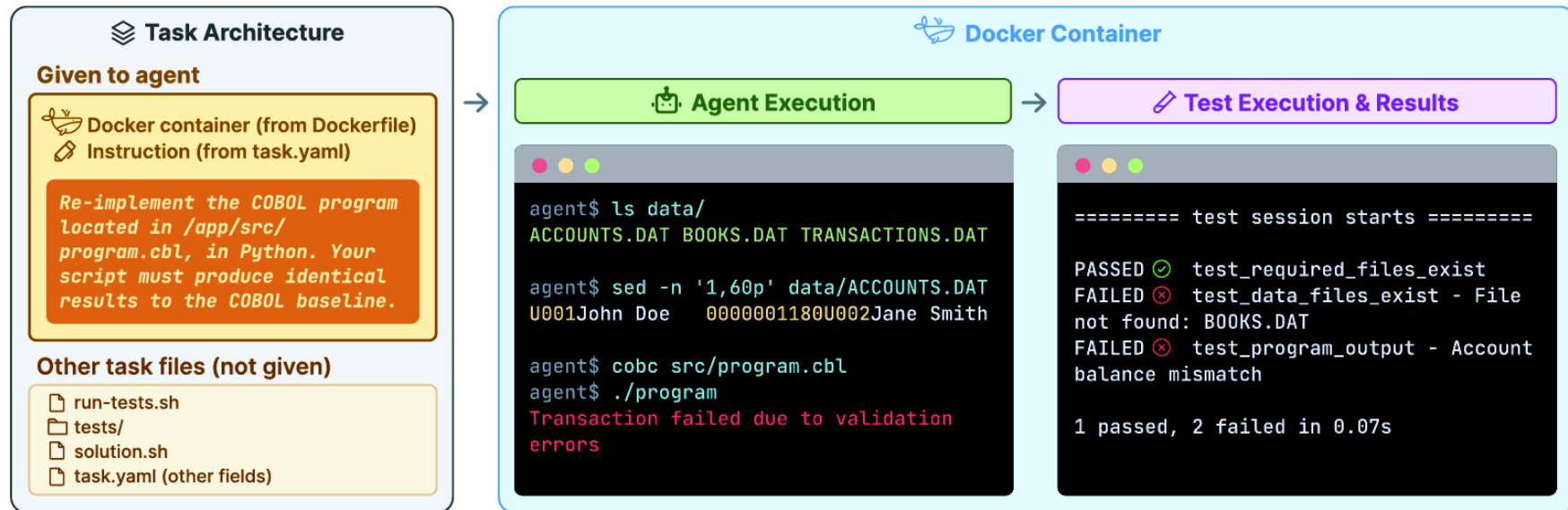


Figure 1: **OpenThoughts3 outperforms existing SFT reasoning datasets across data scales.** All models are finetuned from Qwen-2.5-7B-Instruct. We compare to large SFT datasets (AM, Nemotron Nano) and small curated datasets (s1.1, LIMO) on AIME 2025 (left), LiveCodeBench 06/24-01/25 (middle), and GPQA Diamond (right). Scaling curves for all evaluation benchmarks are in Figure 8.



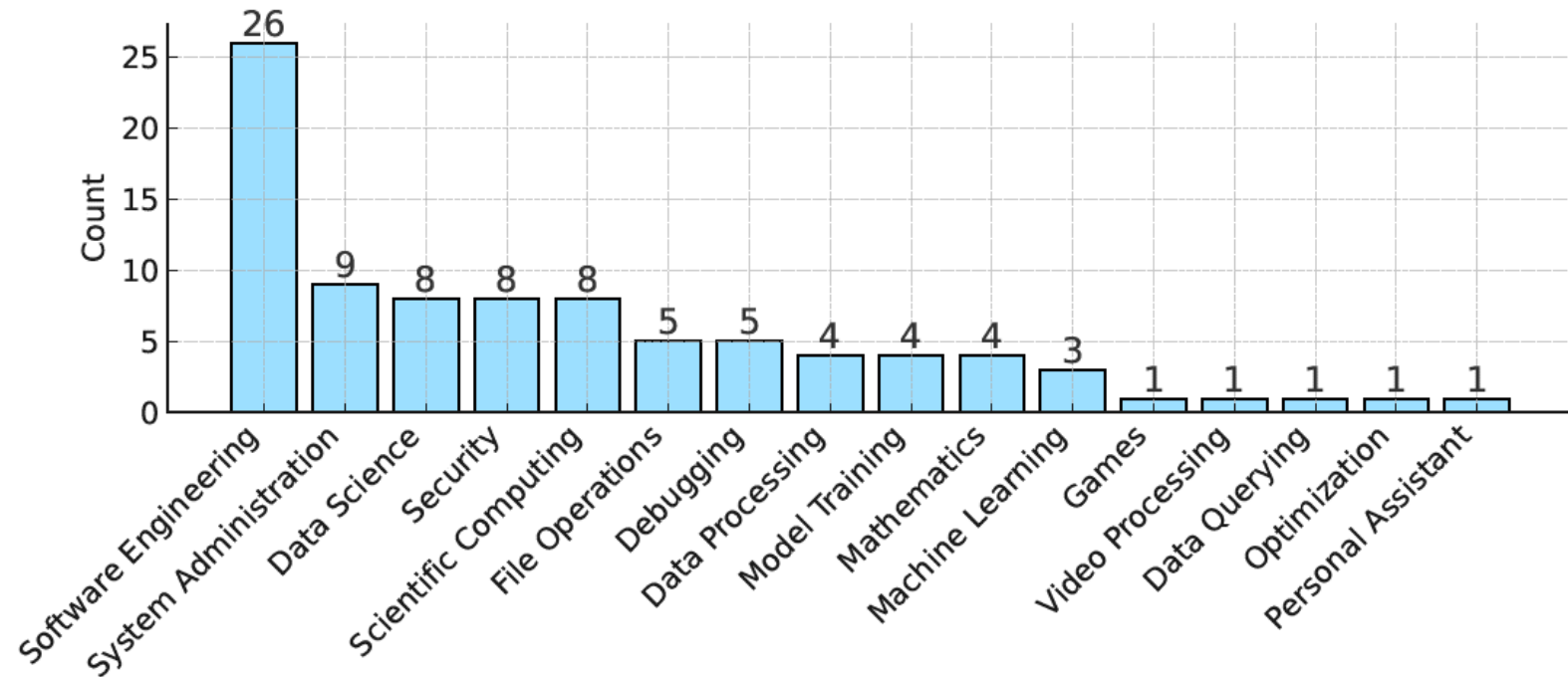
Open foundation models & datasets

- Reproducible evaluation to measure model capabilities: agentic models, Terminal-Bench



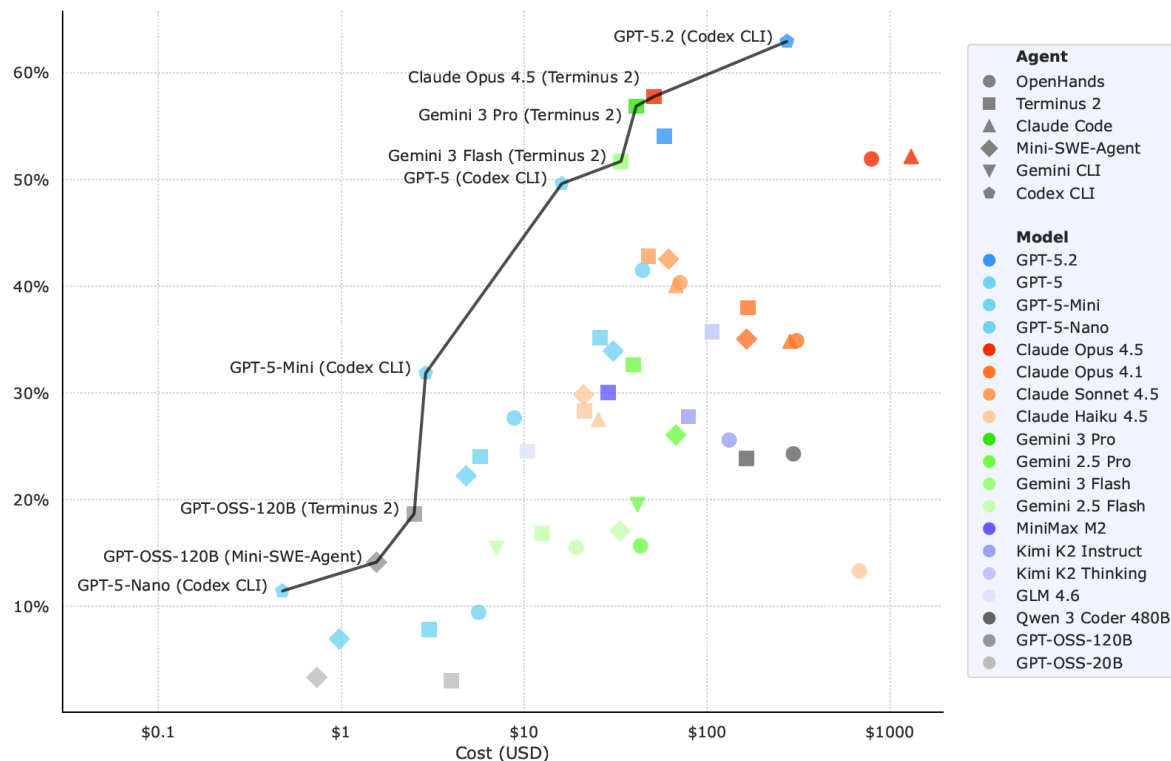
Open foundation models & datasets

- Reproducible evaluation to measure model capabilities: agentic models, Terminal-Bench



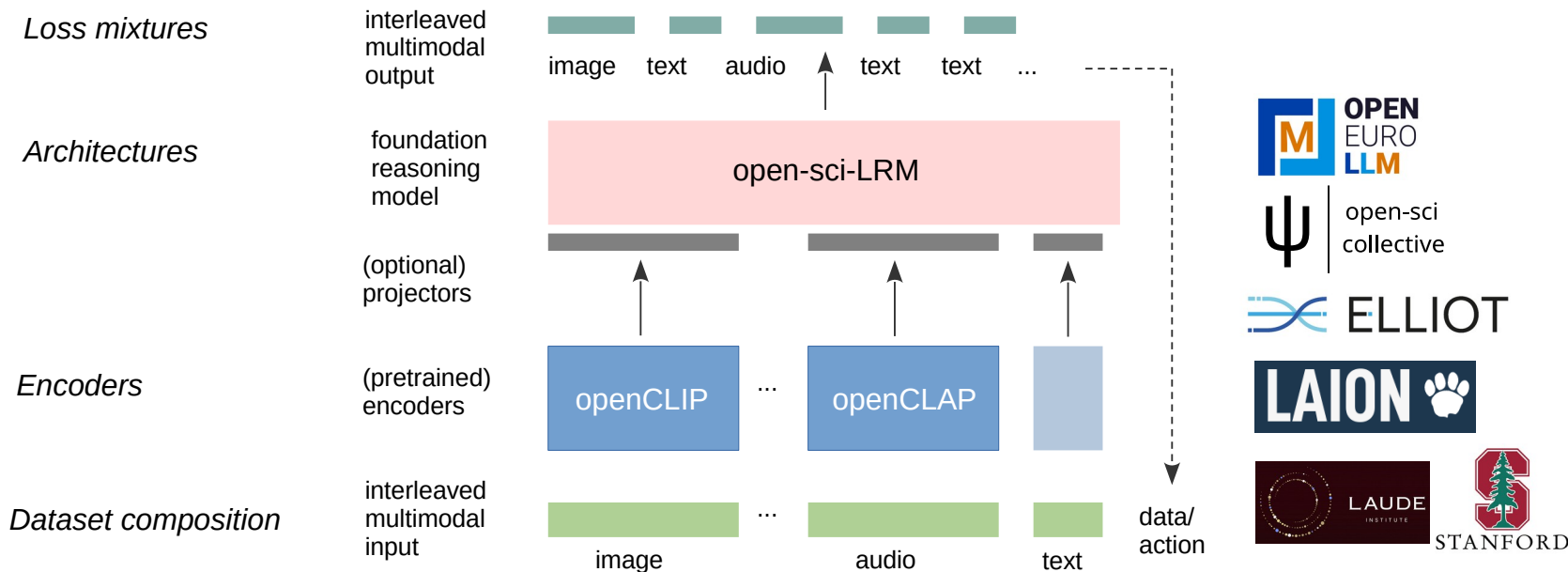
Open foundation models & datasets

- Reproducible evaluation to measure model capabilities: Terminal-Bench



Open foundation models with stronger scalability

- Open pipeline & scaling laws to guide search for stronger scalable open FoMos
- Comparing novel procedures against common reference via scaling laws
- Measurable progress towards stronger agentic systems (Terminal-Bench)



Open foundation models: outlook

- „Moonshot“: **open-sci-MMA - open multi-modal foundation agentic models**
 - Open pipeline, scaling law derivation based search for stronger models
 - Scalable reinforcement learning and test time training
- **OpenEuroLLM, ELLIOT – LAION/ELLIS & friends** : EU consortia for building open foundation models with strongly improved generalization & reasoning
 - Will provide strong multi-modal components for open-sci-MMA
 - Acquired strategic EuroHPC allocation for 3 years – gives some ground to start with, but more is required (esp. dedicated partitions)



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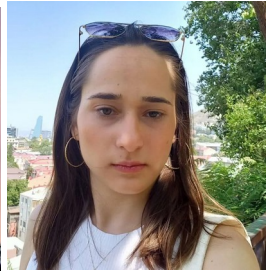
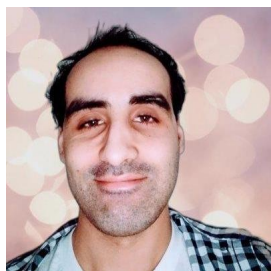
ELLIOT



Tübingen AI Center



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Dr. Mehdi Cherti, Marianna Nezhurina,
JSC



Visit <https://laion.ai/>
Join public LAION and open-sci
Discord server for more projects
and research tracks
> 30k members !

LAION community & friends (Romain Beaumont, Ross Wightmann, Irina Rish, ...)



Prof. Ludwig Schmidt, Stanford



Christoph Schumann

**Let's build open, robust, safe
AI foundations together!**



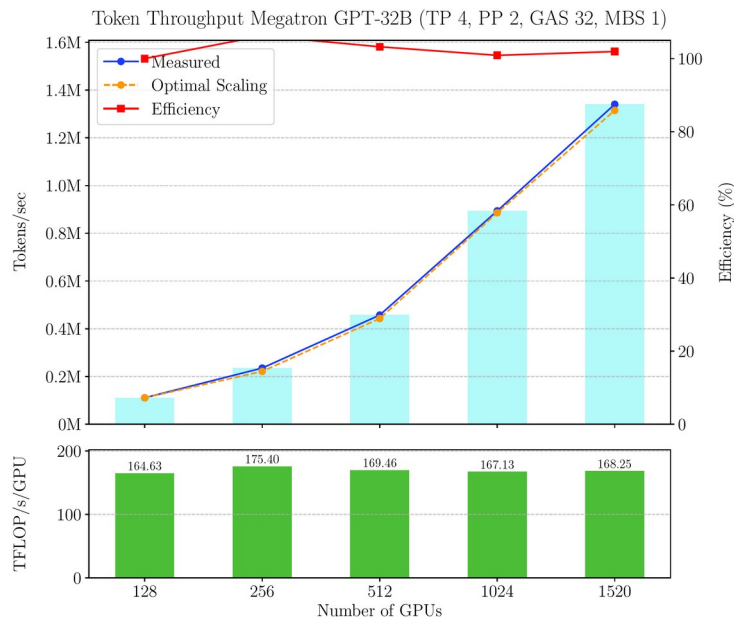
Thanks
for
your
Attention



Supplementary Material

Supercomputers for distributed training

- Distributed training on supercomputers requires scalable code: MegaTron, openCLIP – testing and improving scalable training



Nodes	GPUs	Global BS	Tokens/Step	s/Step	TFLOP/s/GPU	Tokens/s	Efficiency (%)
32	128	512	2,097,152	18.941	164.63	110,722	100.0
64	256	1024	4,194,304	17.830	175.40	235,234	106.2
128	512	2048	8,388,608	18.348	169.46	457,195	103.2
256	1024	4096	16,777,216	18.773	167.13	893,673	100.9
380	1520	6080	24,903,680	18.582	168.25	1,340,238	101.9

Figure 3: Throughput scalability of a 32B parameter GPT pretraining on 32 to 380 nodes on JUWELS Booster using MegaTron-LM, see also Suppl. Tab. 4. GPU utilization (A100 40GB) and token throughput achieve high numbers across various node configurations.



Supercomputing: necessary for progress

- **Compute**: publicly funded supercomputers are of great value for open-source ML/AI community – if strong labs are organized to collaborate
- JUWELS Booster/JURECA (GCS grants, JARA grants)
 - openCLIP (NeurIPS Outstanding Paper Award 2022, > 100M downloads HF, > 10k stars open source repository), DataComp (NeurIPS 2023 Oral), DataComp-LM (NeurIPS 2024), scaling laws works (CVPR 2023, NeurIPS Highlight 2024, ICLR 2025, NeurIPS 2025)
- Oak Ridge Summit (INCITE DoE grant)
 - RedPajama, RedPajama-INCITE (with TogetherAI, COLM 2025)
- Leonardo, JUWELS Booster (EuroHPC Extreme Scale, GCS grants)
 - openThoughts (ICLR 2026 Oral), scaling laws for robust comparison of open foundation models & datasets/openMaMMUT (NeurIPS 2025), Terminal-Bench (ICLR 2026)
- JUPITER
 - openThoughts-Agent (GCS Gauss AI Competition), Terminal-Bench

